

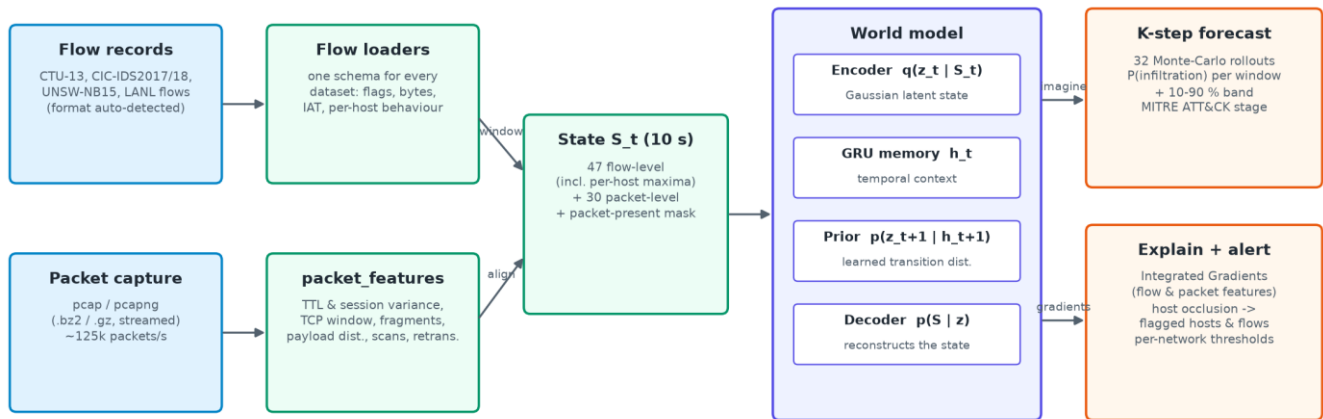
VIGHNAX — BhaviṣyAdvaktā: World-Model Forecasting of Network Attack Progression

Architecture document · SIH 2026 · PS SIH26153 (NTRO) · AI-based Network Attack Forecasting from Network Traffic Data · Team VIGHNAX (Team ID 127364) · open source (MIT), fully offline

1. Idea

Instead of classifying flows one at a time, VIGHNAX learns **how the state of a network evolves** — the transition distribution $P(S_{t+1} | S_t)$ — from flow-level and packet-level telemetry, then rolls that model forward K steps to estimate whether the current trajectory leads to compromise, which MITRE ATT&CK stage comes next, and which traffic features drive the prediction. One model is trained on all seven datasets named in the PS.

2. System architecture



Offline dashboard: live timeline, forecast band, stage mix, attributions, flagged hosts & flows, incident report | no cloud APIs

3. Components

- **Ingestion.** A universal loader reads every PS dataset's native format by content: CTU-13 Argus binetflow, CIC-IDS2017/2018 CICFlowMeter CSV, UNSW-NB15 records, LANL flows; packet captures (pcap/pcapng, .gz/.bz2) are streamed at ~125k packets/s with headers decoded directly (a Scapy reference parser is included).
- **Network state S_t .** Every 10 s window becomes a 78-dim vector: 47 flow-level features (5-tuple statistics, flags, bytes, packets, duration, IAT, fwd/bwd ratios, per-host maxima so one compromised host is not diluted) + 30 packet-level features (TTL mean and per-session variance, TCP window, IP fragment/DF flags, payload-size distribution, SYN-only/RST/ICMP shares, retransmissions, sequential-vs-randomised port-scan signatures) + a packet-present mask.
- **World model.** Encoder $q(z_t | S_t) \rightarrow$ Gaussian latent; GRU memory h_t ; prior $p(z_{t+1} | h_{t+1}) =$ learned transition distribution; decoder $p(S | z)$. Heads on (h, z) give $P(\text{infiltration})$ and a 7-way ATT&CK stage distribution. Loss = reconstruction + $KL(\text{posterior} || \text{prior})$ + attack/stage losses on observed states and on imagined K -step rollouts against the true future labels (supervised dynamics learning from each dataset's attack timeline). Modality dropout (CSV-only inputs), volume augmentation (network size) and dataset-balanced sampling (each epoch draws equally from all seven datasets).
- **Forward simulation and explanations.** 32 Monte-Carlo latent trajectories rolled $K = 6$ windows (60 s) ahead: per-window $P(\text{infiltration})$ with a 10–90 % band and the predicted stage mix with technique IDs. Integrated Gradients (Aumann–Shapley / SHAP family) attributes each forecast to flow- and packet-level features; leave-one-host-out occlusion flags the responsible hosts and flows.
- **Interface.** Offline Flask dashboard (bundled Chart.js and fonts). One drop zone for any mix of flow files, PCAPs or a .zip; the world model streams its forecast window by window; the timeline, forecast fan, attributions and flagged hosts update live at a selectable playback speed ($1\times =$ real time). Alert thresholds are calibrated per network (site profile).

4. Data and evaluation protocol

CTU-13 (flows + full packet captures), CIC-IDS2017 and CSE-CIC-IDS2018 (CICFlowMeter flows), UNSW-NB15 (flow records), LANL cyber1 (router flows + red-team events), DARPA 1999 (tcpdump + attack truth list) and CICIoT2023 (per-activity packet captures). **Every test source is unseen in training:** other botnet families (CTU-13), other days (CIC, UNSW, DARPA), another 12 h slice (LANL), other attack types (CICIoT2023). Validation = every 5th 5-minute block of the training sources. Labels map to ATT&CK stages by behaviour; all seven stages occur. The LR baseline uses the same state features and targets; operating thresholds are chosen per dataset on validation only (most detection with at most 5 % false alarms, same rule for both models).

5. Results on unseen test data (AUROC, world model / LR baseline)

Dataset (unseen test data)	Forecast WM / LR	Early warning WM / LR	Detection WM / LR
CTU-13	0.83 / 0.60	0.82 / 0.48	0.89 / 0.74
CIC-IDS2017	0.56 / 0.56	0.47 / 0.47	0.65 / 0.65
CSE-CIC-IDS2018	0.83 / 0.60	0.57 / 0.55	0.83 / 0.62
UNSW-NB15	0.95 / 0.99	0.63 / 0.71	1.00 / 0.95
LANL cyber1	0.37 / 0.56	0.34 / 0.57	0.40 / 0.55
DARPA 1999	0.63 / 0.69	0.60 / 0.61	0.69 / 0.81
CICIoT2023	n/a	n/a	0.99 / 0.76
Macro average (temporal)	0.70 / 0.67	0.57 / 0.57	0.74 / 0.72

- **Forecasting:** world model ahead on 2 of 6 temporal datasets (macro AUROC 0.70 vs 0.67; LR ahead on UNSW-NB15, LANL cyber1, DARPA 1999); 54 of 201 attack onsets forecast before they began, vs 17 for LR. On validation the world model leads on every dataset; unseen days generalise less well.
- **Stated plainly:** detection of the current window: LR is better on LANL cyber1, DARPA 1999. CICIoT2023 episodes are composed (benign capture then attack capture) and scored for detection only. Thresholds must be calibrated per site; PCAP-only input relies on flows rebuilt from packets.
- **Cost:** 0.04 ms per 10 s window batched on GPU, ~8 ms streamed on a laptop CPU (32 rollouts); ~125k packets/s ingestion. Reproduce: `fetch_datasets.sh` → `prepare_data.py` / `prepare_all.py` → `train.py` → `evaluate.py` (seeded config).

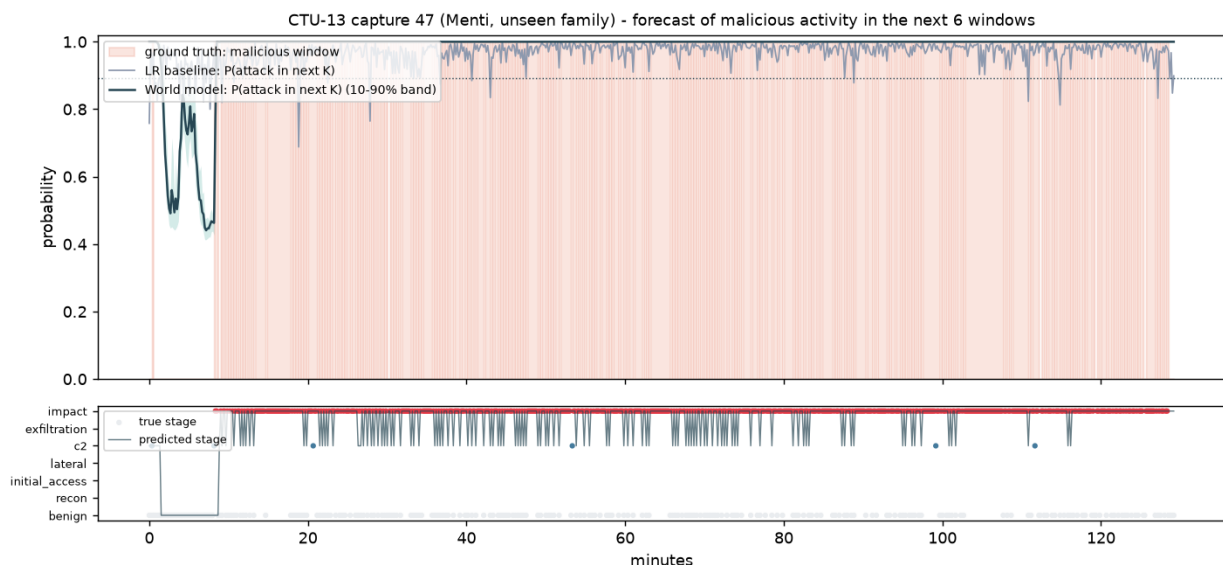


Figure: unseen CTU-13 Menti capture. Shaded = ground-truth malicious windows; dark = world-model $P(\text{attack within } 60 \text{ s})$ with 10–90 % band; grey = LR baseline on the same features.